Math 12 Curriculum

Intermediate Algebra | Middlesex School Mathematics | OpenStax: OpenStax Viewer

COURSE AT A GLANCE

Review Content

REVIEW

Primary: 1.1 Use the Language of Algebra; 1.3 Fractions; 1.5 Properties of Real Numbers
Supplemental: 1.2 Integers

UNIT LEARNING OBJECTIVES

I can evaluate numerical expressions using the order of operations.

I can perform operations with signed integers and fractions.

I can simplify an algebraic expression by combining like terms.

I can use the distributive property to rewrite an expression.

I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

Primary: 2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications
Supplemental: 2.2 Use a Problem Solving Strategy

UNIT LEARNING OBJECTIVES

I can solve a linear equation and present equivalent steps clearly.

I can solve a linear equation containing fractions.

I can rearrange a formula to isolate a specified variable.

I can translate a verbal relationship into a linear equation.

I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

Primary: 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

UNIT LEARNING OBJECTIVES

I can solve and graph a one-variable linear inequality.

I can solve a compound inequality involving AND.

I can solve a compound inequality involving OR.

I can express an inequality solution using interval notation.

I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

Primary: 3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions

UNIT LEARNING OBJECTIVES

I can determine and interpret the slope of a line.

I can graph a line from an equation in a common form.

I can write an equation of a line from given information.

I can determine whether two lines are parallel, perpendicular, or neither.

I can evaluate a function using function notation.

I can determine whether a relation is a function using the vertical line test or input-output data.

I can determine domain and range from a graph or table.

I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

Primary: 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations

UNIT LEARNING OBJECTIVES

I can interpret the solution of a linear system as an intersection point.

I can solve a two-variable linear system by graphing.

I can solve a two-variable linear system by substitution.

I can solve a two-variable linear system by elimination.

I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

Primary: 5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

UNIT LEARNING OBJECTIVES

I can simplify expressions using integer exponent rules.

I can add and subtract polynomials.

I can multiply polynomials, including binomials.

I can divide a polynomial by a monomial.

I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

Primary: Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities; Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities; Intermediate Algebra 2e 6.2 Factor Trinomials; Precalculus 2e 1.2 Domain and Range

UNIT LEARNING OBJECTIVES

I can solve and graph an absolute-value inequality.

I can graph a system of linear inequalities and identify its solution region.

I can factor a quadratic trinomial.

I can identify a relationship as discrete or continuous.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

OpenStax coverage: **Primary:** 1.1 Use the Language of Algebra; 1.3 Fractions; 1.5 Properties of Real Numbers
Supplemental: 1.2 Integers

M12-REV-001

I can evaluate numerical expressions using the order of operations.

OpenStax: **Primary:** 1.1 Use the Language of Algebra

SUPPORTING SKILLS

R Apply the order of operations to a numerical expression.
SK-NUM-ORDER-OPERATIONS · inherited from middle school

M12-REV-002

I can perform operations with signed integers and fractions.

OpenStax: **Primary:** 1.3 Fractions
Supplemental: 1.2 Integers

SUPPORTING SKILLS

R Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school

R Add, subtract, multiply, and divide signed integers.
SK-NUM-INTEGER-OPERATE · inherited from middle school

M12-REV-003

I can simplify an algebraic expression by combining like terms.

OpenStax: **Primary:** 1.1 Use the Language of Algebra

SUPPORTING SKILLS

- R Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- R Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school

M12-REV-004

I can use the distributive property to rewrite an expression.

OpenStax: **Primary:** 1.5 Properties of Real Numbers

SUPPORTING SKILLS

- R Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school

M12-REV-005

I can determine the prime factorization of a whole number.

OpenStax: **Primary:** 1.1 Use the Language of Algebra

SUPPORTING SKILLS

- R Express a whole number as a product of prime factors.
  SK-NUM-PRIME-FACTOR · inherited from middle school

Linear Equations and Applications

REQUIRED

OpenStax coverage: **Primary:** 2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications
Supplemental: 2.2 Use a Problem Solving Strategy

M12-EQN-001

I can solve a linear equation and present equivalent steps clearly.

OpenStax: **Primary:** 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

- I — Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION
- A — Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- I — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS
- A — Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school

M12-EQN-002

I can solve a linear equation containing fractions.

OpenStax: **Primary:** 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

- A — Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- I — Clear fractions from an equation using a common denominator.
  SK-ALG-CLEAR-FRACTIONS
- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A — Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school

M12-EQN-003

I can rearrange a formula to isolate a specified variable.

OpenStax: **Primary:** 2.3 Solve a Formula for a Specific Variable

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I — Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE

M12-EQN-004

I can translate a verbal relationship into a linear equation.

OpenStax: **Primary:** 2.2 Use a Problem Solving Strategy

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I — Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION

M12-EQN-005

I can solve and interpret a number, mixture, or motion problem using a linear equation.

OpenStax: **Primary:** 2.4 Solve Mixture and Uniform Motion Applications
Supplemental: 2.2 Use a Problem Solving Strategy

SUPPORTING SKILLS

- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

Inequalities and Absolute-Value Equations REQUIRED

OpenStax coverage: **Primary:** 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

M12-INE-001

I can solve and graph a one-variable linear inequality.

OpenStax: **Primary:** 2.5 Solve Linear Inequalities

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE
- I Reverse an inequality when multiplying or dividing by a negative value.
  SK-INE-REVERSE-SIGN

M12-INE-002

I can solve a compound inequality involving AND.

OpenStax: **Primary:** 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Interpret AND as the intersection of two inequality solution sets.
  SK-INE-COMPOUND-AND
- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12
- A Reverse an inequality when multiplying or dividing by a negative value.
  SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-003

I can solve a compound inequality involving OR.

OpenStax: **Primary:** 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Interpret OR as the union of two inequality solution sets.
  SK-INE-COMPOUND-OR
- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12
- A Reverse an inequality when multiplying or dividing by a negative value.
  SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-004

I can express an inequality solution using interval notation.

OpenStax: **Primary:** 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12
- I Read inequality symbols and state the relationship each symbol expresses.
  SK-NOT-INEQUALITY-READ
- I Translate between inequality and interval notation.
  SK-NOT-INTERVAL

M12-INE-005

I can solve an absolute-value equation and check its solutions.

OpenStax: **Primary:** 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

- I Rewrite an absolute-value equation as two cases.
  SK-ABS-SPLIT-EQUATION
- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

Lines and Introductory Functions REQUIRED

OpenStax coverage: **Primary:** 3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions

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M12-LIN-001

I can determine and interpret the slope of a line.

OpenStax: **Primary:** 3.2 Slope of a Line

SUPPORTING SKILLS

- I Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH
- I Calculate slope from two points.
  SK-LIN-SLOPE-POINTS
- I Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS

M12-LIN-002

I can graph a line from an equation in a common form.

OpenStax: **Primary:** 3.1 Graph Linear Equations in Two Variables

SUPPORTING SKILLS

- I Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT
- I Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE
- I Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT

M12-LIN-003

I can write an equation of a line from given information.

OpenStax: **Primary:** 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M12-LIN-004

I can determine whether two lines are parallel, perpendicular, or neither.

OpenStax: **Primary:** 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- I Recognize that distinct parallel lines have equal slopes.
  SK-LIN-PARALLEL-SLOPE
- I Determine the negative reciprocal of a nonzero slope.
  SK-LIN-PERPENDICULAR-SLOPE

M12-LIN-005

I can evaluate a function using function notation.

OpenStax: **Primary:** 3.5 Relations and Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT
- I Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ
- I Recall and correctly use the terms relation, function, input, output, domain, and range.
  SK-FUNC-VOCABULARY

M12-LIN-006

I can determine whether a relation is a function using the vertical line test or input-output data.

OpenStax: **Primary:** 3.5 Relations and Functions

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE
- A Recall and correctly use the terms relation, function, input, output, domain, and range.
  SK-FUNC-VOCABULARY · first introduced in Math 12

M12-LIN-008

I can determine domain and range from a graph or table.

OpenStax: **Primary:** 3.5 Relations and Functions

SUPPORTING SKILLS

- I Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH
- I Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE

M12-LIN-009

I can interpret domain and range in context.

OpenStax: **Primary:** 3.5 Relations and Functions

SUPPORTING SKILLS

- I Restrict domain to values meaningful in a context.
  SK-FUNC-CONTEXT-DOMAIN
- I Interpret the range of a function in a context.
  SK-FUNC-CONTEXT-RANGE
- I Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN

Systems of Linear Equations REQUIRED

OpenStax coverage: **Primary:** 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations

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M12-SYS-001

I can interpret the solution of a linear system as an intersection point.

OpenStax: **Primary:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- I Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION

M12-SYS-002

I can solve a two-variable linear system by graphing.

OpenStax: **Primary:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- A Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- I Check an ordered pair in both equations of a system.
  SK-SYS-CHECK
- D Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M12-SYS-003

I can solve a two-variable linear system by substitution.

OpenStax: **Primary:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- I Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE

M12-SYS-004

I can solve a two-variable linear system by elimination.

OpenStax: **Primary:** 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- I Add equations to eliminate a variable.
  SK-SYS-ELIMINATE

M12-SYS-005

I can create and solve a two-variable linear system for an application.

OpenStax: **Primary:** 4.2 Solve Applications with Systems of Equations

SUPPORTING SKILLS

- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- A Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- I Define two variables and write two equations for a context.
  SK-SYS-MODEL
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

Polynomial Operations and Basic Factoring REQUIRED

OpenStax coverage: **Primary:** 5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

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M12-POL-001

I can simplify expressions using integer exponent rules.

OpenStax: **Primary:** 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- I Apply the power rule for exponents.
  SK-EXP-POWER
- I Apply the product rule for exponents.
  SK-EXP-PRODUCT
- I Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT

M12-POL-002

I can add and subtract polynomials.

OpenStax: **Primary:** 5.1 Add and Subtract Polynomials

SUPPORTING SKILLS

- A Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school
- I Identify like polynomial terms by variable part and exponent.
  SK-POLY-LIKE-TERMS
- I Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY

M12-POL-003

I can multiply polynomials, including binomials.

OpenStax: **Primary:** 5.3 Multiply Polynomials

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- I Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY
- I Multiply a monomial by a polynomial.
  SK-POLY-DISTRIBUTE
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-004

I can divide a polynomial by a monomial.

OpenStax: **Primary:** 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- I Divide each term of a polynomial by a monomial.
  SK-POLY-DIVIDE-MONOMIAL
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-005

I can factor a greatest common factor from a polynomial.

OpenStax: **Primary:** 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- I Factor the greatest common monomial factor from a polynomial.
  SK-FACTOR-GCF
- I Determine the numerical and variable greatest common factor of polynomial terms.
  SK-POLY-GCF
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary:** Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities; Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities; Intermediate Algebra 2e 6.2 Factor Trinomials; Precalculus 2e 1.2 Domain and Range

M12-EXT-001

I can solve and graph an absolute-value inequality.

OpenStax: **Primary:** 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

- I Rewrite an absolute-value inequality as a compound inequality.
  SK-ABS-SPLIT-INEQUALITY
- A Interpret AND as the intersection of two inequality solution sets.
  SK-INE-COMPOUND-AND · first introduced in Math 12
- A Interpret OR as the union of two inequality solution sets.
  SK-INE-COMPOUND-OR · first introduced in Math 12
- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12

M12-EXT-002

I can graph a system of linear inequalities and identify its solution region.

OpenStax: **Primary:** 4.7 Graphing Systems of Linear Inequalities

SUPPORTING SKILLS

- I Graph the boundary line of a linear inequality with the correct line type.
  SK-INE-GRAPH-BOUNDARY
- I Identify the overlapping solution region of multiple linear inequalities.
  SK-INE-SYSTEM-REGION
- A Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12

M12-EXT-003

I can factor a quadratic trinomial.

OpenStax: **Primary:** 6.2 Factor Trinomials

SUPPORTING SKILLS

- I Verify a factorization by multiplication.
  SK-FACTOR-CHECK
- I Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL
- I Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC

M12-EXT-004

I can identify a relationship as discrete or continuous.

OpenStax: **Primary:** 1.2 Domain and Range

SUPPORTING SKILLS

- I Distinguish discrete inputs from values varying continuously over an interval.
  SK-FUNC-DISCRETE-CONTINUOUS